Multiplication of algebraic fractions

1 Simplify the following expressions:

a $\dfrac{1}{a} \times \dfrac{1}{b}$

b $\dfrac{a}{3} \times \dfrac{a}{7}$

c $\dfrac{b}{q} \times \dfrac{k}{u}$

d $\dfrac{c}{3} \times \dfrac{d}{2}$

e $\dfrac{3y}{8} \times \dfrac{4y}{9}$

f $\dfrac{9x}{2} \times \dfrac{5y}{7}$

g $\dfrac{4x}{5} \times \dfrac{3y}{7}$

h $\dfrac{5u}{3a} \times \dfrac{3v}{5b}$

i $\dfrac{11s}{7t} \times \dfrac{3r}{5q}$

j $\dfrac{9a}{10b} \times \dfrac{3b}{8a}$

k $\dfrac{8x}{7y} \times \dfrac{9y}{5x}$

l $\dfrac{9u}{7v} \times \dfrac{8w}{11y}$

m $\dfrac{14u}{15v} \times \dfrac{40v}{24q}$

n $\dfrac{9u}{77v} \times \dfrac{110v}{24q}.$

o $\dfrac{x^2}{3} \times \dfrac{6}{x}$

p $\dfrac{p^2}{q} \times \dfrac{q^2}{p}$

q $\dfrac{r^2}{n} \times \dfrac{n^2}{r}$

r $\dfrac{5y^2}{9} \times 27y^2$

s $\dfrac{5y}{2} \times 6y^5.$

t $\dfrac{4m^2}{20} \times \dfrac{3y^2}{7m}$

u $\dfrac{7a^2}{15b^2} \times \dfrac{10b}{11a}.$

v $\dfrac{11a^2}{4b^2} \times \dfrac{10b}{3a}$

w $\left(-\dfrac{10}{21x}\right) \times \dfrac{3x^2}{110}$

x $\dfrac{2n^2}{6} \times \dfrac{-2y^2}{3n}$

2 Simplify:

a $\dfrac{m}{3} \times \dfrac{5m}{9} \times 81q$

b $\dfrac{2n}{3m} \times \dfrac{21q}{4n} \times \dfrac{10m}{49p}$

3 Simplify $\left(\dfrac{x}{3y}\right)^2 \times \dfrac{5x^2}{2y}.$

4 Complete the following:

$$\frac{\square}{3a^2} \times \frac{7b}{\square} = \frac{14bc}{9a^2d}$$

5 The product of two fractions is 1. If one of the fractions is $\dfrac{3x}{4yz}$, what is the other fraction?

Division of algebraic fractions

6 Simplify the following expressions:

a $\dfrac{r}{3} \div \dfrac{2}{m}$

b $\dfrac{u}{3} \div \dfrac{4}{v}$

c $\dfrac{m}{4} \div \dfrac{m}{3}$

d $\dfrac{m}{28} \div \dfrac{23}{20}$

e $\dfrac{3y}{6} \div \dfrac{4y}{7}$

f $\dfrac{2x}{9} \div \dfrac{7}{5y}$

g $\dfrac{3y^2}{4} \div \dfrac{12}{29y^6}$

h $\dfrac{4u}{35y} \div \dfrac{14q}{50y}$

i $\dfrac{-2x}{11} \div \dfrac{2x}{3}.$

j $\dfrac{9u}{36v} \div \dfrac{7v}{36u}$

k $\dfrac{-9n}{4} \div \dfrac{11n}{8}.$

l $\dfrac{u^2}{12} \div \dfrac{u}{9}$

m $\dfrac{4x^2y}{9z^4} \div \dfrac{6xy^5}{15z}$

n $\dfrac{u^2}{9} \div \dfrac{u}{6}.$

o $\dfrac{8u^3}{32v} \div \dfrac{5v^3}{96u}$

p $\dfrac{3w^3}{7} \div 9w^6$

q $\dfrac{16u}{17v} \div (-14uv)$

r $\dfrac{45x^2yz}{10} \div \dfrac{9y^2}{5x}$

s $\dfrac{35t^2u^2}{12v^2w^2} \div \dfrac{35t^3u}{12w^2y^2}$

t $\dfrac{10t^2u^3}{35v^3w^3} \div \dfrac{10t^3u^2}{20w^3y}$

7 Simplify $\left(\dfrac{2x}{y}\right)^4 \div \left(\dfrac{z}{3x^2y}\right)^2.$

8 Simplify $\dfrac{3xy}{12y} \times \dfrac{4y^2}{3wx} \div \dfrac{15wx}{3w^2}.$

9 Complete the following:

$$\frac{\square}{4p^2} \div \frac{2q^2}{3r} = \frac{27r^3}{\square}$$

10 A rectangle has an area of $\dfrac{5x^3y^4}{3pq}$ and a length of $\dfrac{4xy}{p}$. Find an expression for the width of the rectangle.